



Crash Course in Maths for Cognitive Science

Session 2 | Statistics & Inference

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September 2026

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What today is about

Last session we studied *single* uncertain quantities. Today we turn to *data*: many observations at once, and what we can learn from them.

- Summarise data with a few numbers: *mean, variance, covariance*.
- Understand why averaging works: the *Law of Large Numbers* and the *Central Limit Theorem*.
- Infer from a sample to a population: *standard errors, confidence intervals, hypothesis tests*.
- Build our first real model: *linear regression*.

As in Session 1, we stop to *solve examples together* at every step.

The thread running through everything

Population vs. sample

There is a fixed but unknown truth about the whole *population* (for example, the true average reaction time of all humans). We only ever see a limited *sample*. Statistics is the disciplined art of guessing the population from the sample, and honestly reporting how uncertain that guess is.

Keep asking, on every slide: is this a statement about the population, or about our particular sample?

The centre of a distribution: expectation

The first thing we want to know about data is where it is centred.

Definition

The **expectation** (mean) is the probability-weighted average value:

$$\mathbb{E}[X] = \sum_x x p(x) \quad (\text{discrete}), \quad \mathbb{E}[X] = \int x f(x) dx \quad (\text{continuous}).$$

Picture the distribution as a pile of mass on the number line: the expectation is its *balance point*. It need not be a value the variable can actually take, a fair die has mean 3.5.

Expectation adds up nicely

The property that makes expectation so useful is *linearity*, and it holds always, even when variables influence each other.

Useful fact

$$\mathbb{E}[aX + bY + c] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c.$$

Intuition

This is why we can rescale, shift, and combine measures freely and still track their average, no need to re-derive the whole distribution.

Worked example 1: rescaling a score

Example

A fair die X is rolled, and a game pays $Y = 2X + 1$ points.

1. What is $\mathbb{E}[X]$?
2. What is the expected payout $\mathbb{E}[Y]$?

Try it yourself first — the solution is on the next slide.

Worked example 1: solution

1. $\mathbb{E}[X] = \frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = \frac{21}{6} = 3.5$.
2. By linearity, $\mathbb{E}[Y] = 2\mathbb{E}[X] + 1 = 2(3.5) + 1 = 8$, no need to average the six payouts separately.

Linearity turns a distribution-level question into one line of arithmetic. It will reappear constantly (e.g. mean-centring data subtracts $\mathbb{E}[X]$ from every value).

How spread out is the data? Variance

Definition

The **variance** is the average squared distance from the mean; the **standard deviation** is its square root (back in the original units):

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2], \quad \text{SD}(X) = \sqrt{\text{Var}(X)}.$$

We square the deviations so positive and negative ones do not cancel, and to penalise large departures more. The square root brings us back to sensible units, milliseconds, not milliseconds-squared.

Three variance rules worth memorising

Useful fact

Computational formula: $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$, usually fastest by hand.

Shift and scale: $\text{Var}(aX + b) = a^2 \text{Var}(X)$, adding a constant does not change spread; scaling multiplies the SD by $|a|$.

Sum: $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y)$.

That last term, the covariance, is exactly the correction for when X and Y move together. It vanishes only when they are uncorrelated.

Worked example 2: variance of a die

Example

Find the variance and standard deviation of a fair die, using the computational formula $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$.

Try it yourself first — the solution is on the next slide.

Worked example 2: solution

The mean of the square:

$$\mathbb{E}[X^2] = \frac{1}{6}(1 + 4 + 9 + 16 + 25 + 36) = \frac{91}{6} \approx 15.17.$$

So

$$\text{Var}(X) = 15.17 - 3.5^2 = 15.17 - 12.25 = 2.92, \quad \text{SD}(X) = \sqrt{2.92} \approx 1.71.$$

A die roll typically lands about 1.7 away from its mean of 3.5.

Worked example 3: spread of a neuron's firing

Example

A neuron's firing rate (spikes/s) is measured on five trials:

6, 8, 10, 12, 14.

Compute the mean, the variance (average squared deviation), and the standard deviation.

Try it yourself first — the solution is on the next slide.

Worked example 3: solution

Mean = $\frac{6+8+10+12+14}{5} = 10$. Deviations $-4, -2, 0, 2, 4$; their squares $16, 4, 0, 4, 16$ sum to 40. So

$$\text{Var} = \frac{40}{5} = 8, \quad \text{SD} = \sqrt{8} \approx 2.83.$$

Intuition

The SD (≈ 2.8 spikes/s) is in the same units as the data, so it is directly interpretable, unlike the variance ($8 \text{ spikes}^2/\text{s}^2$). This is why error bars and “noise levels” are almost always quoted as SDs.

How two variables move together

Definition

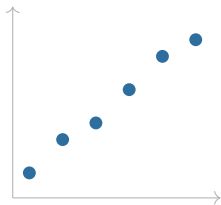
The **covariance** measures whether two quantities tend to be large or small together:

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)],$$

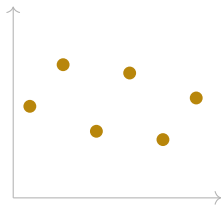
and the **correlation** rescales it into $[-1, 1]$: $\rho = \frac{\text{Cov}(X, Y)}{\text{SD}(X) \text{SD}(Y)}.$

Covariance is positive when the two are usually on the same side of their means, negative when one is high while the other is low.

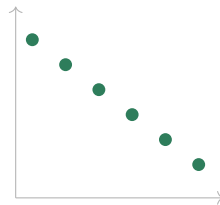
What correlation looks like



$$\rho \approx +0.9$$



$$\rho \approx 0$$



$$\rho \approx -0.9$$

Correlation measures the strength and direction of a *linear* trend: $+1$ a perfect rising line, 0 no linear trend, -1 a perfect falling line.

Useful facts about correlation

Useful fact

- ρ is *unit-free* and always lies in $[-1, 1]$.
- It is unchanged by shifting or positively rescaling either variable (mm or cm, °C or °C+5 give the same ρ).
- $\rho = \pm 1$ exactly when the points lie on a perfect straight line.

Key idea

Collecting all pairwise covariances of many variables gives the **covariance matrix**, the single object at the heart of PCA (Session 4). A statistical quantity becomes a geometric one.

Worked example 4: do two neurons co-vary?

Example

Across five trials, neuron A 's rate is $x = (1, 2, 3, 4, 5)$ and neuron B 's rate is $y = (2, 4, 5, 4, 5)$ (arbitrary units); the means are $\bar{x} = 3$, $\bar{y} = 4$.

Compute the correlation ρ between the two neurons. (*Helpful sums to find: $\sum(x - \bar{x})(y - \bar{y})$, $\sum(x - \bar{x})^2$, $\sum(y - \bar{y})^2$.*)

Try it yourself first — the solution is on the next slide.

Worked example 4: solution

The deviation products give

$$\sum (x - \bar{x})(y - \bar{y}) = 6, \quad \sum (x - \bar{x})^2 = 10, \quad \sum (y - \bar{y})^2 = 6.$$

So

$$\rho = \frac{6}{\sqrt{10 \times 6}} = \frac{6}{\sqrt{60}} \approx 0.77 :$$

a strong positive association, the two neurons tend to be high or low together (a *noise correlation*). We reuse these very sums for the regression line at the end of today.

Worked example 5: guess the sign

Example

For each pair, is the correlation *positive*, *negative*, or *roughly zero*?

1. Hours of sleep vs. reaction time (more sleep $\rightarrow$ faster responses).
2. Stimulus contrast vs. a neuron's firing rate.
3. A person's height vs. their exam score.

Try it yourself first — the solution is on the next slide.

Worked example 5: solution

1. **Negative:** more sleep $\rightarrow$ *lower* reaction time.
2. **Positive:** higher contrast $\rightarrow$ higher firing rate.
3. **Roughly zero:** no reason for a linear link.

Common pitfall

$\rho = 0$ means no *linear* relationship, not “no relationship”, points on $y = x^2$ have $\rho = 0$ yet are perfectly predictable. And correlation is never, by itself, evidence of *causation*: a strong ρ is an invitation to investigate, not a conclusion about cause.

Practice — try it

A variable X takes the values $\{2, 4, 6\}$, each with probability $\frac{1}{3}$.

1. Find $\mathbb{E}[X]$.
2. Find $\text{Var}(X)$ using the computational formula.
3. If $Y = 3X$, what is $\text{SD}(Y)$?

Answers. (1) $\mathbb{E}[X] = 4$. (2) $\text{Var}(X) = \frac{56}{3} - 16 = \frac{8}{3} \approx 2.67$. (3) $\text{SD}(Y) = 3 \text{SD}(X) = \sqrt{24} \approx 4.90$.

Two theorems behind all of statistics

Why can we trust an average of noisy measurements? Two results answer this.

Key idea

Law of Large Numbers: as the sample size n grows, the sample mean $\bar{X}_n$ converges to the true mean μ . *Averages settle down.*

Central Limit Theorem: for large n , the sample mean is approximately Normal, $\bar{X}_n \approx \mathcal{N}(\mu, \sigma^2/n)$, *whatever the shape of the original data. Averages become bell-shaped.*

The Central Limit Theorem, in pictures



$n = 1$ (one die)



$n = 2$ (average of two)



large n (bell-shaped)

One die is *flat*; the average of two is already *triangular*, average enough and the shape is *Normal*, no matter what you started with.

Why the Central Limit Theorem is remarkable

The Law of Large Numbers is intuitive: more data, better estimate. The Central Limit Theorem says something stronger.

Even if individual observations are wildly non-Normal, a skewed reaction time, a 0/1 correct response, their *average* traces out a smooth Normal curve, with spread shrinking like $\sigma/\sqrt{n}$.

Link to cognitive science

This underwrites *trial averaging* in EEG and fMRI: averaging n noisy trials cuts the noise SD by $\sqrt{n}$, pulling a buried event-related signal out of the noise. It also explains why so many measured traits look Gaussian, each is a sum of many small independent contributions.

The $\sqrt{n}$ law of averaging

Useful fact

If $X_1, \dots, X_n$ are independent with mean μ and SD σ , the sample mean $\bar{X}$ has

$$\mathbb{E}[\bar{X}] = \mu, \quad \text{SD}(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

The key consequence is a *diminishing return*: to *halve* the uncertainty of an estimate you need *four times* as much data; to cut it tenfold, a hundredfold. This is why large studies are expensive.

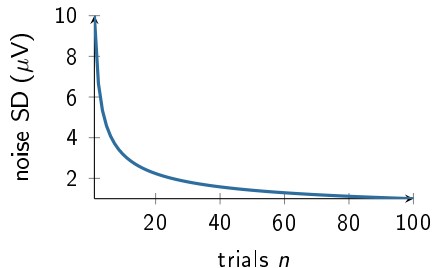
Worked example 6: how many EEG trials?

Example

Single-trial EEG responses carry noise with SD $\sigma = 10 \mu\text{V}$.

1. What is the noise SD after averaging 25 trials?
2. How many trials are needed to bring it down to $1 \mu\text{V}$?

Try it yourself first — the solution is on the next slide.



Worked example 6: solution

1. $SD = \frac{10}{\sqrt{25}} = \frac{10}{5} = 2 \mu V.$
2. Solve $\frac{10}{\sqrt{n}} = 1 \Rightarrow \sqrt{n} = 10 \Rightarrow n = 100$, four times as many trials as the 25 that gave $2 \mu V.$

Every halving of the noise costs a *quadrupling* of trials, the practical face of the $\sqrt{n}$ law, and why long recording sessions have sharply diminishing returns.

Definition

A **parameter** is a fixed unknown describing the population (e.g. the true mean μ). An **estimator** is a recipe applied to the sample (e.g. the sample mean $\bar{X}$). It is **unbiased** if it is right *on average* across many samples: $\mathbb{E}[\hat{\theta}] = \theta$.

Why we divide by $n - 1$

Common pitfall

The sample variance divides by $n - 1$, not n : $s^2 = \frac{1}{n - 1} \sum_{i=1}^n (x_i - \bar{x})^2$.

Using the sample mean $\bar{x}$ (instead of the unknown true mean) makes the deviations come out slightly too small; dividing by $n - 1$ corrects exactly for this, giving an unbiased estimate. Know both the rule and the reason.

Standard deviation vs. standard error

This distinction is subtle and constantly misused, so it gets its own slide.

Common pitfall

The **standard deviation** s describes the spread of the raw data; it does *not* shrink with more data. The **standard error** $SE = s/\sqrt{n}$ describes the precision of our *estimate of the mean*; it *does* shrink as n grows.

When a figure shows error bars, always ask which they are: the spread of individuals, or the uncertainty in the mean? They tell very different stories.

Worked example 7: which one shrinks?

Example

A population of reaction times has SD $\sigma = 20$ ms.

1. Compute the standard error of the mean for a sample of $n = 25$.
2. And for $n = 100$.
3. As n grows from 25 to 100, what happened to the SD of the data, and to the SE of the mean?

Try it yourself first — the solution is on the next slide.

Worked example 7: solution

1. $SE = \frac{20}{\sqrt{25}} = 4 \text{ ms.}$

2. $SE = \frac{20}{\sqrt{100}} = 2 \text{ ms.}$

3. The *SD* stays 20 ms (it is a property of the data, not the sample size). The *SE* halved (from 4 to 2 ms): we know the mean twice as precisely with four times the data.

Reporting uncertainty: confidence intervals

Definition

A 95% confidence interval for the mean is $\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}}$.

The width $1.96 \sigma / \sqrt{n}$ comes straight from the Central Limit Theorem and the 95% mark of the Normal curve. The recipe is built so that, across many repeated samples, 95% of the intervals it produces contain the true μ .

When σ is unknown: use t , not z

Useful fact

In practice we estimate σ by the sample SD s . This extra uncertainty means the standardised mean follows *Student's t -distribution* with $n - 1$ degrees of freedom, like the Normal, but with heavier tails.

For small samples the 95% multiplier exceeds 1.96 (about 2.26 at $n = 10$), and it shrinks toward 1.96 as n grows. Rule of thumb: once $n \gtrsim 30$, the difference hardly matters.

Worked example 8: a confidence interval for reaction time

Example

Over $n = 64$ trials the mean reaction time is $\bar{x} = 540$ ms; assume the population SD is $\sigma = 80$ ms.

1. Give the 95% confidence interval for the true mean (use 1.96).
2. Give the 99% interval (use 2.576). Which is wider, and why?

Try it yourself first — the solution is on the next slide.

Worked example 8: solution

The standard error is $SE = \frac{80}{\sqrt{64}} = \frac{80}{8} = 10$ ms.

- 95%: $540 \pm 1.96(10) = 540 \pm 19.6 = [520.4, 559.6]$ ms.
- 99%: $540 \pm 2.576(10) = 540 \pm 25.8 = [514.2, 565.8]$ ms.

The 99% interval is *wider*: demanding more confidence means casting a wider net.
Confidence and precision trade off.

What “95% confidence” really means

Common pitfall

A 95% confidence interval does *not* mean “there is a 95% probability the true mean lies in *this* interval”. In the classical framework μ is fixed; it is the interval that is random.

The correct reading is about the *procedure*: repeat the experiment many times and 95% of the intervals you build this way would capture the truth.

Practice: inference

Practice — try it

1. A sample of $n = 100$ has mean $\bar{x} = 52$ with known $\sigma = 10$. Give a 95% confidence interval for μ .
2. Testing $H_0 : \mu = 50$ against $\mu \neq 50$, what is the z-statistic and the rough conclusion?
3. A treatment raises scores by 2 points, with within-group SD 10. What is Cohen's d , and is it small, medium, or large?

Answers. (1) $SE = 1$, CI $[50.0, 54.0]$. (2) $z = 2.0$, reject H_0 at 5%. (3) $d = 0.2$, small.

The logic of a hypothesis test

Four steps

- (1) State a **null hypothesis** H_0 (“nothing is going on”) and an alternative H_1 .
- (2) Choose a **test statistic** that would be extreme if H_1 were true.
- (3) Compute the **p -value**: assuming H_0 , the probability of a result at least this extreme.
- (4) If $p < \alpha$ (often 0.05), **reject** H_0 .

Worked example 9: is the coin fair?

Example

A coin lands heads 60 times in 100 flips. Test $H_0 : p = 0.5$ against $p \neq 0.5$.

Hint: under H_0 the number of heads has mean $np = 50$ and standard deviation $\sqrt{np(1-p)} = \sqrt{25} = 5$.

Try it yourself first — the solution is on the next slide.

Worked example 9: solution

The standardised statistic:

$$z = \frac{60 - 50}{5} = 2.0.$$

A result at least this extreme (either tail) has probability $\mathbb{P}(|Z| \geq 2.0) \approx 0.046$. Since $0.046 < 0.05$, we **reject** H_0 : weak but statistically significant evidence of bias.

Worked example 10: is the subject above chance?

Link to cognitive science

Testing whether accuracy beats chance is the most common test in psychophysics.

Example

A subject is correct on 44 of 60 trials in a yes/no task where chance is 0.5. Under H_0 the number correct has mean 30 and SD $\sqrt{60 \cdot 0.5 \cdot 0.5} = \sqrt{15} \approx 3.87$. Compute the z-statistic and state the conclusion.

Try it yourself first — the solution is on the next slide.

Worked example 10: solution

$$z = \frac{44 - 30}{3.87} = \frac{14}{3.87} \approx 3.6.$$

This is far beyond 1.96 ($p < 0.001$), so the subject performs *well above chance*. A large z here reflects both a real effect and a decent number of trials.

Definition

A **Type I error** rejects a true H_0 (a false positive), with probability α . A **Type II error** fails to reject a false H_0 (a false negative), with probability β . The **power** is $1 - \beta$, the chance of detecting a real effect.

There is always a trade-off: making α very small (few false positives) tends to lower power (more missed real effects). Choosing the balance is a design decision, not a mathematical one.

The workhorse tests

All follow the same four-step logic; they differ only in the statistic.

- **t -test**, compares means (one sample, or two groups).
- **χ^2 test**, observed vs. expected counts in categories.
- **ANOVA** (F -test), means across three or more groups.

Each carries assumptions worth stating out loud: independence of observations, and approximate Normality of the relevant averages.

Worked example 11: significant, or just big?

Example

Two groups of $n = 25$ have score means 105 vs. 100, with a pooled SD of 10.

1. Compute Cohen's $d = \frac{\text{mean}_1 - \text{mean}_2}{\text{SD}}$.
2. The standard error of the difference is $\text{SD}\sqrt{1/n_1 + 1/n_2}$. Compute the t -statistic. Is the difference significant at 5%?

Try it yourself first — the solution is on the next slide.

Worked example 11: solution

1. $d = \frac{105 - 100}{10} = 0.5$, a solidly “medium” effect.
2. $SE_{\text{diff}} = 10\sqrt{\frac{1}{25} + \frac{1}{25}} = 10\sqrt{0.08} \approx 2.83$, so $t = \frac{5}{2.83} \approx 1.77$, giving $p \approx 0.08$: *not* significant at 5%.

Intuition

A genuine, medium-sized effect can fail the test purely because the sample is too small. Significance depends on both the effect *and* n , so always report an *effect size* alongside the p -value.

The p -value and the replication crisis

Link to cognitive science

The p -value is routinely misunderstood. It is *not* the probability that H_0 is true, nor that the result was “due to chance”. Misuse of $p < 0.05$, through multiple comparisons, optional stopping, and publication bias, is a central driver of the **replication crisis** that reshaped psychology and cognitive neuroscience since the 2010s.

Reporting effect sizes and confidence intervals, and pre-registering analyses, are part of the modern remedy.

From correlation to prediction: regression

Correlation measures the strength of a linear relationship. Regression draws the actual line and lets us *predict*.

Definition

Simple linear regression fits $\hat{y} = a + bx$ by **least squares**, choosing a, b to minimise $\sum(y_i - \hat{y}_i)^2$. The solution is

$$b = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} = \frac{\text{Cov}(x, y)}{\text{Var}(x)}, \quad a = \bar{y} - b\bar{x}.$$

What least squares is doing

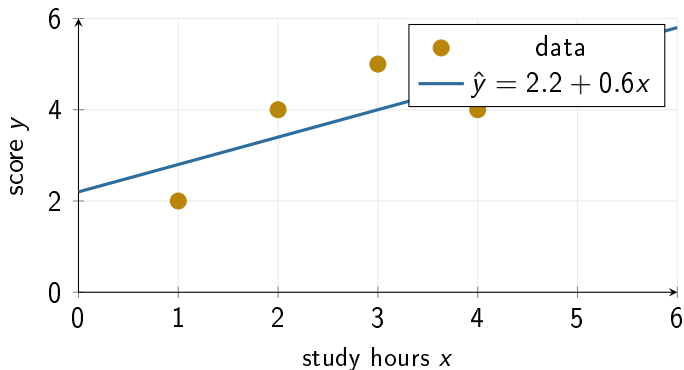
Intuition

We look for the straight line passing as close as possible to all the points, where “close” is the squared vertical gaps. The slope is just the covariance of x and y rescaled by the spread of x , the same covariance from an hour ago.

Two facts that make good checks

The residuals always sum to zero and are uncorrelated with x ; and the line always passes through the mean point $(\bar{x}, \bar{y})$.

The regression line, drawn



The fitted line through the study-hours data. Each extra hour of study predicts about 0.6 more points.

Worked example 12: fit the study-hours line

Example

For the study-hours data ($\bar{x} = 3$, $\bar{y} = 4$) we already found $\sum(x - \bar{x})(y - \bar{y}) = 6$ and $\sum(x - \bar{x})^2 = 10$, with correlation $\rho \approx 0.77$.

1. Find the slope b and intercept a .
2. What fraction of the variance in scores does the line explain?

Try it yourself first — the solution is on the next slide.

Worked example 12: solution

$$b = \frac{6}{10} = 0.6, \quad a = \bar{y} - b\bar{x} = 4 - 0.6(3) = 2.2, \quad \hat{y} = 2.2 + 0.6x.$$

The fraction of variance explained is

$$R^2 = \rho^2 \approx 0.77^2 \approx 0.60,$$

so the line accounts for about 60% of the variation in scores; the remaining 40% is scatter the line cannot capture.

Worked example 13: predicting firing from contrast

Link to cognitive science

Fitting firing rate as a function of a stimulus variable is the simplest *neural encoding model*.

Example

Contrast $x = (0, 1, 2, 3, 4)$ and firing rate $y = (1, 1, 2, 2, 4)$, with $\bar{x} = \bar{y} = 2$. Here $\sum(x - \bar{x})(y - \bar{y}) = 7$ and $\sum(x - \bar{x})^2 = 10$.

1. Give the regression line.
2. Predict the firing rate at contrast $x = 5$.

Try it yourself first — the solution is on the next slide.

Worked example 13: solution

$$b = \frac{7}{10} = 0.7, \quad a = 2 - 0.7(2) = 0.6, \quad \hat{y} = 0.6 + 0.7x.$$

At contrast $x = 5$: $\hat{y} = 0.6 + 0.7(5) = 4.1$ spikes/s.

Common pitfall

This is an *extrapolation* beyond the observed range ($x \leq 4$): trust it only if the linear trend really continues. Real tuning curves eventually saturate, so predicting far outside the data is risky.

Regression is the atom of modeling

Link to cognitive science

Extend regression to many predictors and it becomes *multiple regression*; add a nonlinear “link” and it becomes the *Generalized Linear Model* behind psychometric functions and neural encoding models (Session 5); view it as “fit parameters to minimise error” and it is exactly the objective of supervised machine learning.

And geometrically, least squares is a *projection*, which is where linear algebra picks up the story next session.

Practice — try it

Four points: $x = (0, 1, 2, 3)$, $y = (1, 1, 2, 4)$.

1. Find $\bar{x}$ and $\bar{y}$.
2. Compute $\sum(x - \bar{x})(y - \bar{y})$ and $\sum(x - \bar{x})^2$.
3. Give the regression line $\hat{y} = a + bx$.

Answers. (1) $\bar{x} = 1.5$, $\bar{y} = 2$. (2) $\sum(x - \bar{x})(y - \bar{y}) = 5$, $\sum(x - \bar{x})^2 = 5$. (3) $\hat{y} = 0.5 + x$.

Session 2: what we built

- Summaries: the *mean* (centre), *variance/SD* (spread), and *covariance/correlation* (co-movement).
- The *LLN* and *CLT*: averages converge and become Normal with spread $\sigma/\sqrt{n}$, the basis of standard errors and confidence intervals.
- *Hypothesis testing*: *p*-values, the two errors, and why significance is not importance.
- *Regression*: a least-squares line, reusing covariance and pointing ahead to the GLM and machine learning.

Next: linear algebra, vectors, matrices, and geometry.